UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE & TECHNOLOGY

SCHOOL OF ICT

COMPUTER ENGENEERING

Mobile Application Assignment

Flutter Lab 6

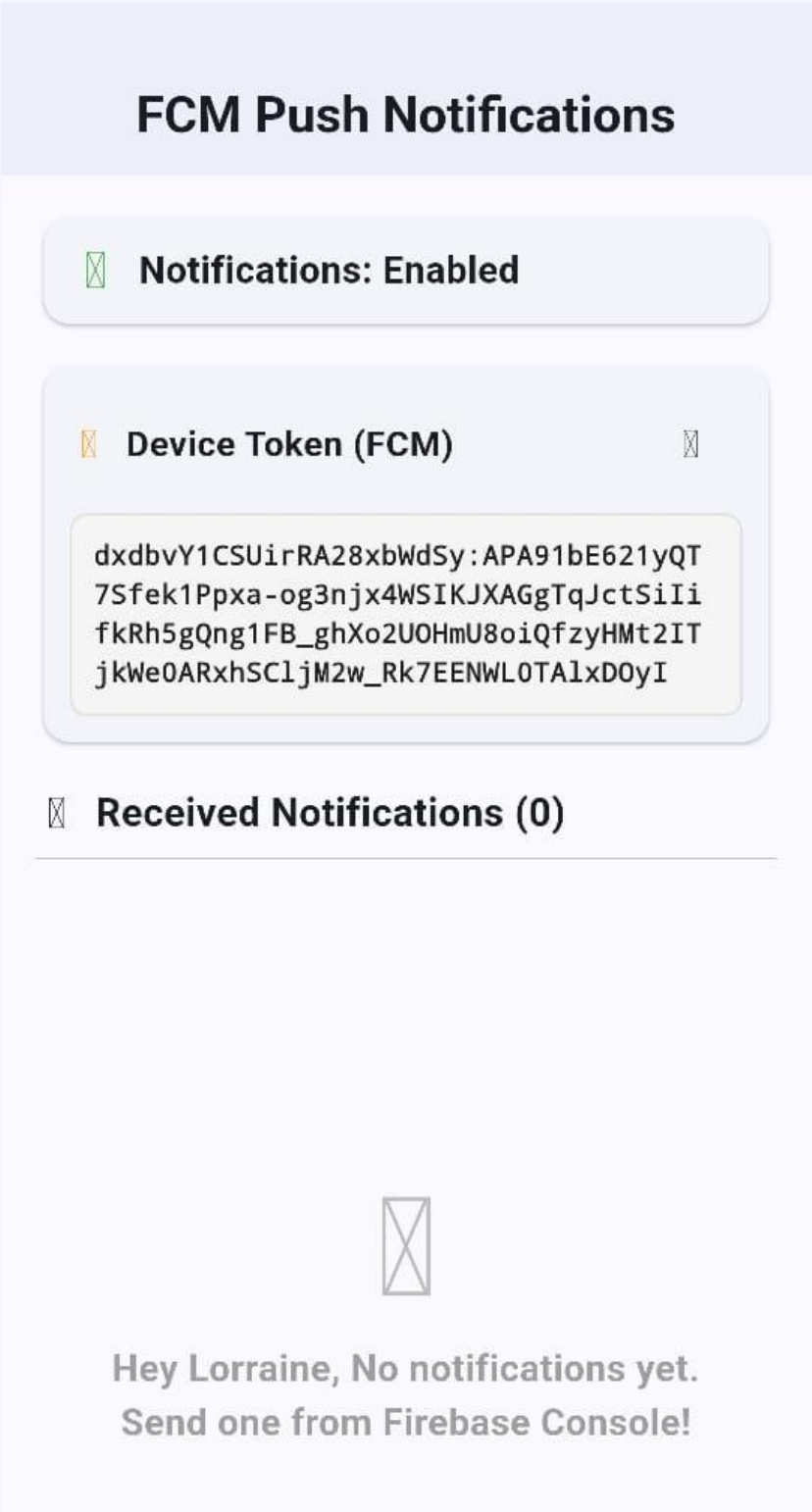
Firebase Cloud Messaging (FCM)

REG NO: 223021337

Section 1 : Screenshots

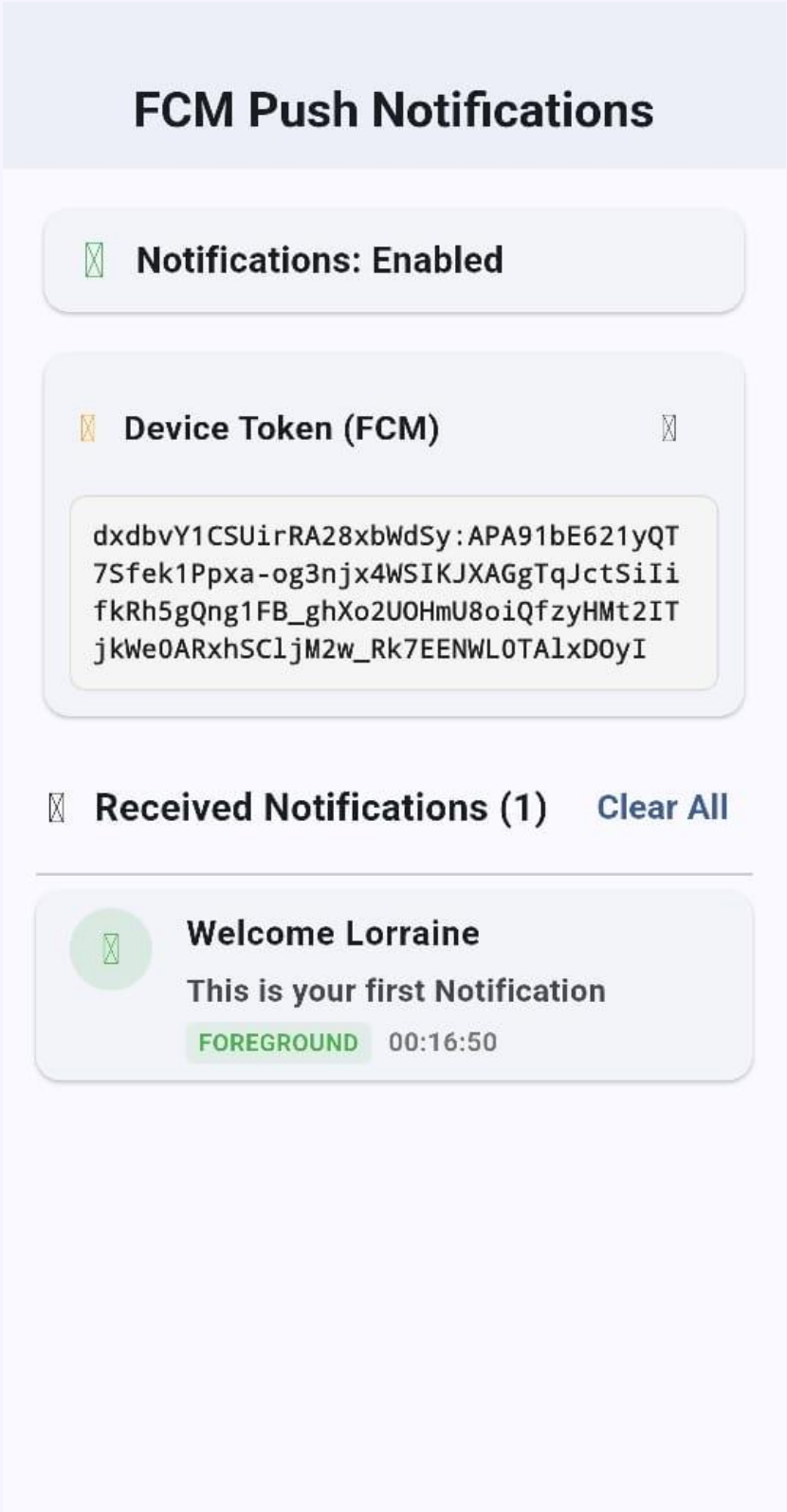
1.1 Home Screen — Notification Feed

The main screen. Shows the device token at the top and a list of all received notifications below. Each card shows the notification title, body, and time received.



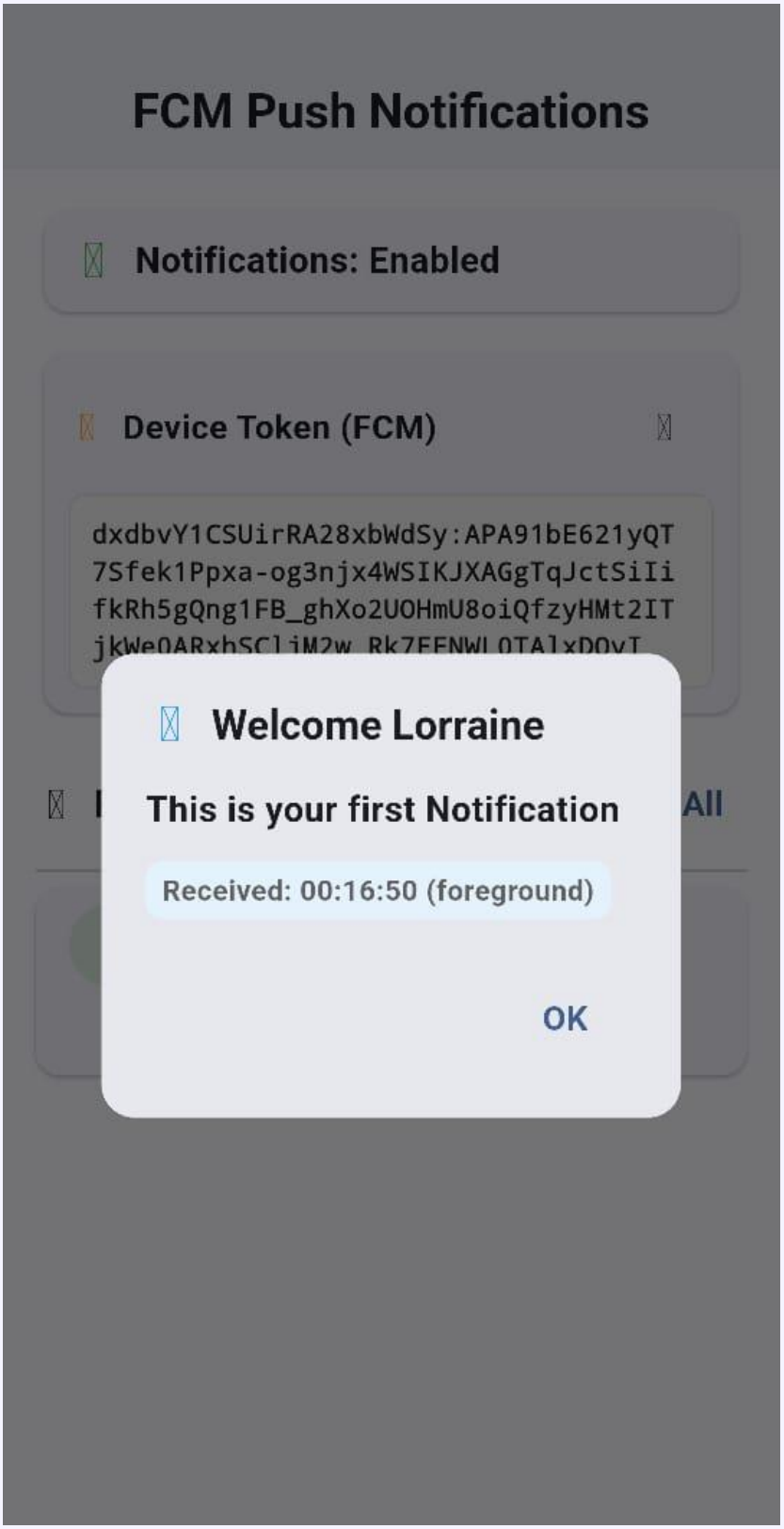
1.2 Received Notification — In-App List

After sending a notification from Firebase Console, it appears in the list on the home screen with the title, body, and time. This shows the message content displayed inside the app UI as required.



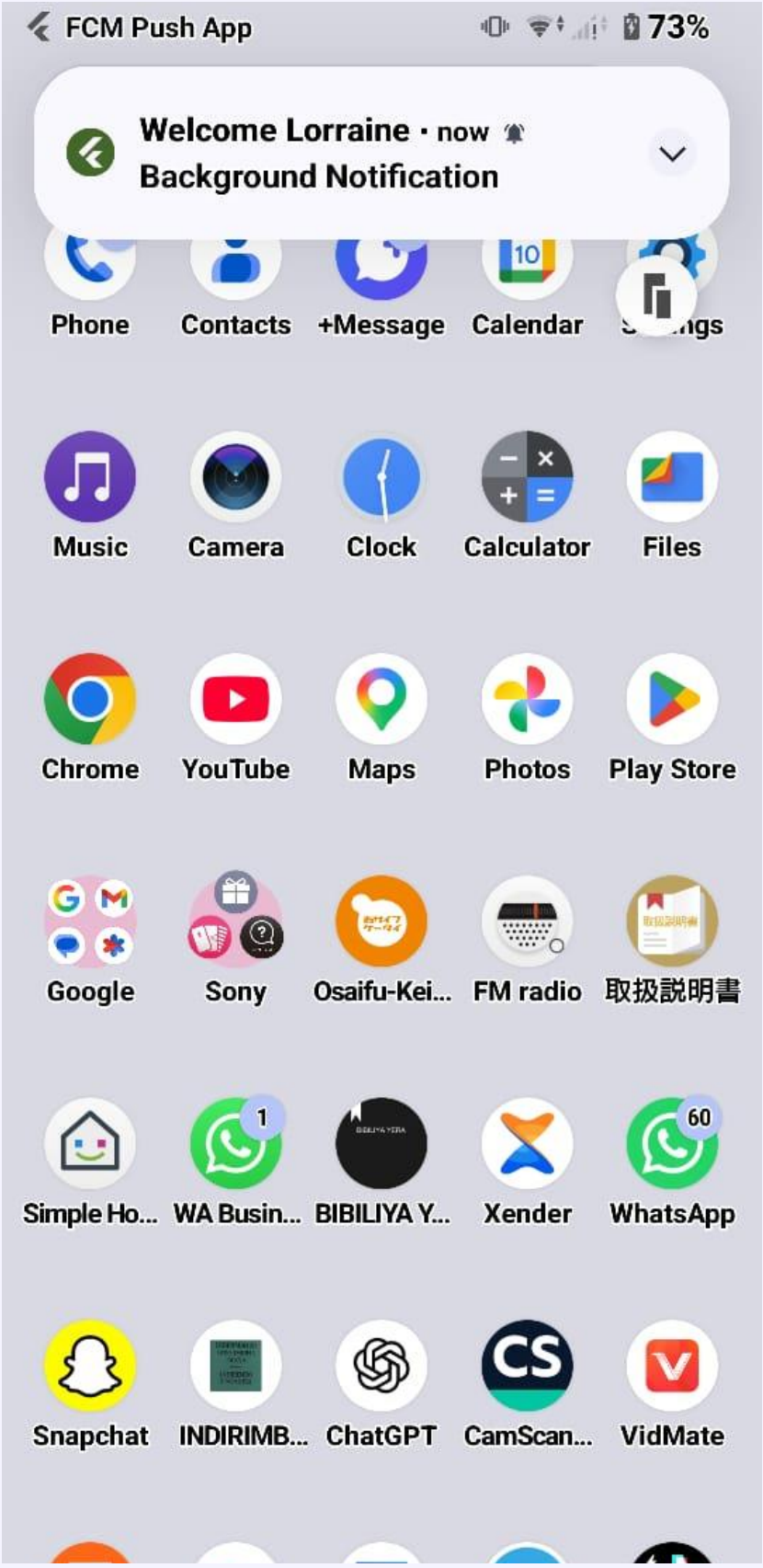
1.3 Notification Popup — App is Open (Foreground)

When the app is open and a notification arrives, an AlertDialog popup appears automatically. It shows the notification title and body and must be dismissed by the user. This satisfies the 'show a popup when notification is received' requirement.



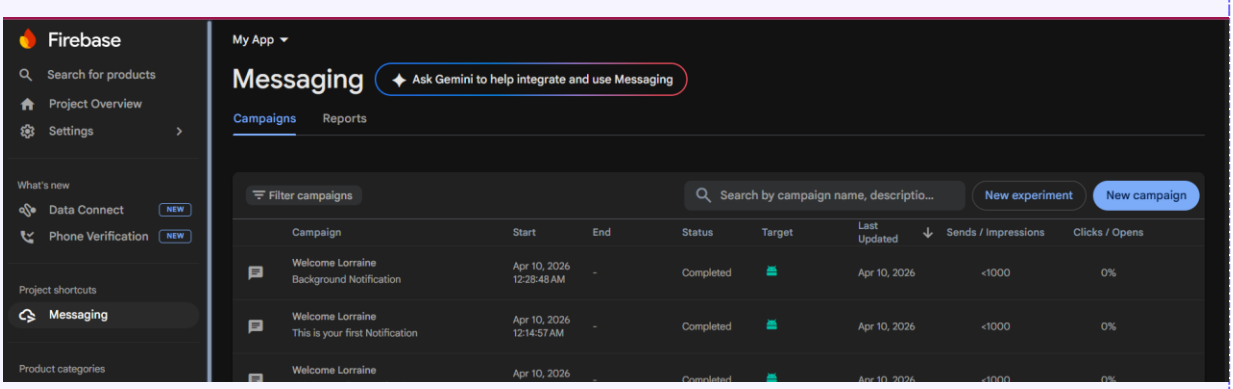
1.4 Background Notification — System Tray

When the app is minimised or the phone screen is on the home screen, the FCM notification appears in the Android notification tray at the top of the screen. Tapping it opens the app and adds the message to the list.



2.2 Firebase Cloud Messaging Console Page

This page shows all my notification campaigns and has the button to create a new notification.



HANDWRITTEN REPORT

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Mobile Application FLutter Lab 6

GENERAL OBSERVATIONS

1. Foreground Behaviour

When the application is open on the emulator, Firebase notifications do NOT automatically appear in the system tray. Instead, they must be manually intercepted using the FirebaseMessaging onMessage stream listener. The app handle this by showing a custom in-app popup banner that slides down from the top of the screen, providing a better user experience than the default system notification.

2. Background Behaviour

When the app is minimised or the screen is on the home screen, FCM notifications are handled automatically by the Firebase SDK and appear directly in the Android system notification tray. Tapping the notification launches the app and the message is captured via FirebaseMessaging onMessagingOpenedApp automatically added to the notification history list.

3. Terminated State (App closed)

When the app is fully closed (not running in the background), notifications still arrive in the system tray. Upon reopening the app, the initial notifications is retrieved using FirebaseMessaging getInitialMessage() and displayed in the history. This ensures notifications are lost even if the app was not running.

4. Device Token Behaviour

Each Android emulator instances generates a unique FCM registration token. The token is fetched asynchronously on app startup and displayed in the device Token tab. The token can change if the app is reinstalled or the emulator is reset. The app listeners for token refresh events via onTokenRefresh to handle this automatically.

5. Emulator vs Real Device

Testing on the Android emulator works correctly for FCM when Google Play Services are enabled on the emulator image. The emulator must use a "Google Play" or "Google APIs" system image — the basic emulator image without Google services will NOT receive FCM notifications. Notification sounds and vibration are not available on the emulator but work on real devices.

6. Notification Persistence

Received notifications are saved using SharedPreferences so they persist across app restarts. The notification history is loaded on startup, meaning the user can always see their notification history even after closing and reopening the app.

Key Steps Followed

Step 1:

Create the Flutter Application

Step 2:

Set Up Firebase Project

Step 3:

Integrate Firebase into Flutter

Step 4:

Configure Firebase Messaging in the App

Step 5:

Retrieve and Display the Device Token

Step 6:

Handle Foreground Notifications (App Open)

Step 7:

Handle Background and Terminated State

Step 8:

Configure Android Notification channel

Step 9:

Send a Test Notification From Firebase Console

Step 10:

Test All App States

CS Scanned with CamScanner